Notes: Caballero and Hammour (AER, 1994)
The Cleansing Effect of Recessions

Contents

1 Big picture

- **Research question.** How do industries that are continuously replacing old production units with new ones respond to cyclical fluctuations in demand? More specifically, do recessions mainly reduce new creation, increase destruction of old units, or both?
- **Core idea.** In an economy with creative destruction, production units embody different vintages of technology. New units are created with the frontier technology, while older, less productive units eventually become obsolete and are destroyed.
- **Two margins.** A fall in demand can be absorbed by reducing creation of new units or by increasing destruction of old units. A representative-firm model cannot separate these two margins, so the paper uses a vintage model with heterogeneous production units.
- **Insulation mechanism.** When creation can adjust cheaply and quickly, a recession can be absorbed by a sharp drop in creation. Existing units are then insulated from the fall in demand, and destruction need not rise.
- **Role of adjustment costs.** When fast creation is costly, firms smooth creation over time. Because creation does not fully absorb demand fluctuations, prices fall in recessions and old units are destroyed more aggressively.
- **Empirical fact.** In U.S. manufacturing job-flow data, job destruction is much more cyclical than job creation. Destruction spikes in recessions, while creation is relatively smoother and more symmetric.
- **Main result.** A calibrated vintage model with modest adjustment costs can match the relative volatility and asymmetry of job creation and job destruction.
- **Economic takeaway.** Recessions can have a "cleansing" effect: they disproportionately eliminate older, less productive, or less profitable production units. This is a positive statement about what happens in downturns, not a normative claim that recessions are desirable.

2 Incremental contribution of the paper

- **Business cycles meet creative destruction.** The paper brings the Schumpeterian idea of ongoing creation and destruction into a business-cycle setting driven by demand fluctuations.
- **Separation of gross flows.** Instead of focusing only on net employment or net investment, the paper studies gross creation and gross destruction separately.
- **A mechanism for asymmetric job flows.** The model explains why job destruction can be sharply countercyclical while job creation is much smoother: creation is a forward-looking, smoothed margin; destruction is tied more directly to current profitability.
- **Insulation as a theoretical benchmark.** The paper shows that with linear creation costs, demand shocks are absorbed entirely by the creation margin. This full-insulation case is useful because it clarifies why convexity or smoothing costs are necessary for destruction to respond.
- **Connection to job-flow data.** The model is taken to the Davis and Haltiwanger manufacturing job-flow series, linking a theoretical vintage model to observed gross job creation and destruction.

- **Calibration discipline.** The paper calibrates the model using manufacturing employment and output data, showing that a simple vintage framework can reproduce nontrivial features of the data.
- **Clarification of the liquidationist view.** The paper rehabilitates the positive idea that recessions cleanse low-productivity units, while explicitly separating this from the old normative liquidationist claim that recessions are good.

3 Model environment and data

3.1 Vintage production units

The model studies an industry with exogenous technical progress. A production unit created at time t_0 embodies the frontier technology available at t_0 and produces a constant flow $A(t_0)$ over its lifetime. The frontier productivity $A(t)$ grows at rate $\gamma > 0$.

- Production units differ by age and therefore by productivity.
- New units embody the newest technology.
- Old units remain in operation until their profits fall to zero.
- Creation costs slow technology adoption, so multiple vintages coexist.

Let $f(a,t)$ be the density of production units of age a at time t, and let $\bar{a}(t)$ be the age of the oldest operating unit. Employment and the active capital stock are proportional to

$$N(t) = \int_0^{\bar{a}(t)} f(a,t)\,da.$$

Industry output is

$$Q(t) = \int_0^{\bar{a}(t)} A(t-a)f(a,t)\,da.$$

Interpretation. The industry is not summarized by a single representative plant. It is a distribution of vintages, and business cycles can change both the inflow of new units and the outflow of old units.

3.2 Creation and destruction

The rate of creation is the boundary value $f(0,t)$: the number of new units entering at time t. The destruction margin is governed by the age cutoff $\bar{a}(t)$. A lower $\bar{a}(t)$ means that younger units are destroyed, so endogenous destruction rises.

Normalized creation and destruction rates are denoted:

$$CC(t) = \frac{f(0,t)}{N(t)}, \qquad DD(t) = \frac{\text{destruction flow}}{N(t)}.$$

The model therefore maps naturally into gross job-flow data: job creation corresponds to the entry of new production units, and job destruction corresponds to the exit of existing production units.

3.3 Market equilibrium

The industry is perfectly competitive and in partial equilibrium. Creating a new unit costs

$$c = c(f(0,t)), \qquad c(\cdot) > 0, \quad c'(\cdot) \geq 0.$$

The dependence of c on the creation rate captures the idea that fast creation may be costly because of installation costs, training costs, capacity constraints in the capital-goods sector, or heterogeneous entry costs.

Key equilibrium conditions are:

- **Free entry.** The cost of creating a new unit equals the present discounted value of its future profits.
- **Free exit.** The oldest unit in operation earns zero profit:

$$P(t)A(t - \bar{a}(t)) = 1.$$

- **Demand.** Total industry spending is exogenous:

$$P(t)Q(t) = D(t).$$

Interpretation. A fall in demand lowers industry revenue. Whether that fall shows up as lower creation or higher destruction depends on how expensive it is to move creation over time.

3.4 Steady state

With constant demand D^*, the industry has a balanced-growth steady state. The lifetime of a production unit is constant, $\bar{a}(t) = \bar{a}^*$, and the age distribution is a truncated exponential:

$$f^*(a) = f^*(0)e^{-\delta a}, \qquad 0 < a < \bar{a}^*.$$

The steady-state lifetime $\bar{a}^*$ balances the benefit of adopting newer technology against the cost of replacing old units. Faster technical progress lowers the efficient lifetime of a unit, while higher creation costs increase the lifetime needed to recover entry costs.

3.5 Data

The empirical application uses U.S. manufacturing job-flow data from Davis and Haltiwanger. These are quarterly series for manufacturing plants, covering the period 1972:2-1986:4. The calibration exercise focuses on 1972:2-1983:4.

- **Job creation.** Gross employment gains at expanding and entering plants.
- **Job destruction.** Gross employment losses at contracting and exiting plants.
- **Output proxy.** The growth rate of industrial production is used as the empirical counterpart to demand-driven output movement.

- **Sector regressions.** The paper uses two-digit SIC manufacturing data and constrains coefficients to be equal across sectors except for constants.

Interpretation. The model is not a full labor-market model. The job-flow data are used because fixed-proportion production units make employment a natural measure of creation and destruction.

4 Mechanism and empirical specifications

4.1 Full insulation with linear creation costs

The clearest benchmark is the case with constant creation cost, $c'(f(0,t)) = 0$. Then the free-entry and free-exit conditions determine the destruction cutoff independently of demand. The cutoff remains at the steady-state age $\bar{a}^*$.

In this full-insulation case:

- demand fluctuations are absorbed by changing $f(0,t)$;
- the destruction age $\bar{a}(t)$ does not respond to demand;
- existing units are insulated from demand fluctuations by the adjustment in creation;
- destruction can still vary because of "echo" effects from past creation, but not because current demand directly changes the cutoff age.

Economic intuition. If creating new units can be adjusted instantly and cheaply, a recession mainly means fewer new projects are started. Old projects do not need to be scrapped more aggressively.

4.2 Partial insulation with convex creation costs

When $c'(f(0,t)) > 0$, fast creation is costly. Firms therefore prefer to smooth creation over time. A recession cannot be fully absorbed by a drop in current creation because that would require very costly catch-up creation later.

The paper uses the linear creation-cost specification

$$c(f(0,t)) = c_0 + c_1 f(0,t), \qquad c_0, c_1 > 0.$$

With positive adjustment costs:

- creation falls in recessions but not enough to absorb the full shock;
- prices fall more than under full insulation;
- old units become unprofitable earlier;
- destruction rises, especially in sharp downturns.

Economic intuition. Creation is a smoothed, forward-looking margin. Destruction is a current-profitability margin. This asymmetry is why destruction reacts more sharply to recessions.

4.3 Regression evidence on job flows

The main descriptive regression relates sectoral job creation and destruction rates to leads, contemporaneous values, and lags of output growth:

$$Y_{it} = \alpha_i + \sum_{j=-2}^{2} \beta_j \widehat{Q}_{i,t+j} + \varepsilon_{it},$$

where Y_{it} is either job creation or job destruction and $\widehat{Q}_{it}$ is the growth rate of industrial production for sector i.

The paper also splits output growth into positive and negative components:

- $\widehat{Q}^{+}$: output growth above its mean, interpreted as expansionary movements;
- $\widehat{Q}^{-}$: output growth below its mean, interpreted as contractionary movements.

This split tests the model's prediction that creation smooths asymmetric demand movements while destruction preserves or amplifies recession asymmetry.

4.4 Calibration exercise

The calibration asks whether the model can reproduce the observed gross job-flow patterns. The paper uses the model in two ways:

- **Employment-driven simulation.** The observed employment path is treated as given. The model splits net employment changes into gross creation and destruction.
- **Output-driven simulation.** The observed output path is used to infer demand movements. The model then generates implied job creation and destruction series.

The model is solved as a periodic equilibrium over the sample period. The adjustment-cost parameter c_1 is chosen to minimize the weighted sum of squared residuals of the model-generated creation and destruction series.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies a mechanism that can rationalize observed gross job-flow dynamics. It does not run a modern causal design in which recessions are randomly assigned. Instead, it uses a structured model and asks whether a vintage creative-destruction mechanism can match salient empirical moments.

The strongest empirical facts are:

- job destruction is much more responsive to output growth than job creation;
- destruction responds especially strongly in contractions;

- creation is much more symmetric and smoother than destruction;
- a calibrated model with modest creation adjustment costs can reproduce these patterns.

5.2 Why creation is smoother

New production units last for many periods. Firms therefore decide creation by averaging expected future demand over the lifetime of the unit. This makes creation relatively smooth and roughly symmetric around its mean.

5.3 Why destruction is asymmetric

Destruction is governed by whether existing units are currently profitable. A sharp fall in demand lowers prices and makes old units unprofitable. Because the old units are already marginal, destruction can spike quickly in recessions.

5.4 Cleansing versus pit-stop views

The paper distinguishes the cleansing view from the pit-stop view of recessions.

- **Cleansing view.** Recessions destroy outdated or unprofitable production units, changing the composition of the productive structure.
- **Pit-stop view.** Recessions are periods when productivity-improving activities are undertaken because their opportunity cost is temporarily low.

Caballero and Hammour emphasize the cleansing view, but they do not deny that other mechanisms may operate simultaneously.

5.5 Limitations

- The model is stylized and assumes fixed proportions of labor and capital inside production units.
- Demand fluctuations are exogenous; the paper does not model the source of recessions.
- The empirical exercise is a calibration and moment-matching exercise rather than a causal microeconometric test.
- The model abstracts from many labor-market frictions, plant-level uncertainty, financing frictions, and endogenous innovation decisions.
- The cleansing effect can be hidden in aggregate productivity data because labor hoarding, externalities, and measurement issues can make average productivity procyclical even when low-productivity units are being destroyed.

6 Tables and figures

6.1 Figure 1: Steady-state cross-sectional density

Read:

- Figure 1 shows the steady-state distribution of production units by age.
- The density declines with age because units depreciate or fail exogenously over time.
- The distribution is truncated at $\bar{a}^*$, the steady-state age of endogenous obsolescence.
- Units older than $\bar{a}^*$ are not operated because their productivity is too low relative to the frontier and the market price.

Economic message: Creative destruction gives the industry a moving age distribution. The oldest vintages are the margin on which recessions can cleanse the productive structure.

6.2 Figure 2: Symmetric demand and partial insulation

Read:

- Figure 2 simulates the model under a symmetric sinusoidal demand fluctuation.
- Panel A shows that creation CC and destruction DD both respond over the cycle.
- Panel B shows the underlying change in demand.
- Because creation costs are convex, creation does not fully insulate existing units. Destruction rises when demand weakens.

Economic message: Once fast creation is costly, the destruction margin becomes cyclical. Recessions are partly absorbed by scrapping old units, not only by reducing new entry.

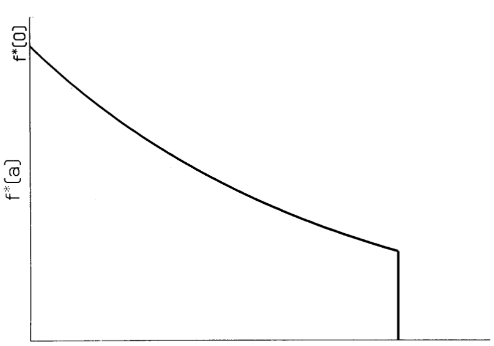

Figure 1: Steady-state cross-sectional density.

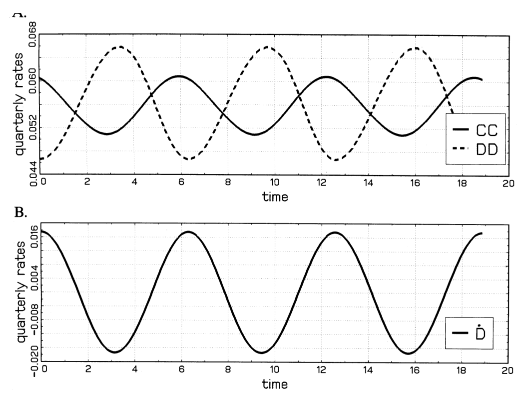

Figure 2: Creation, destruction, and symmetric demand.

6.3 Figure 3 and Table 1: Job-flow evidence

Read Figure 3:

- Panel A plots job creation and job destruction rates in U.S. manufacturing.
- Panel B plots the growth rate of industrial production.
- Job destruction spikes sharply in recessions, especially around the mid-1970s and early 1980s.
- Job creation is much smoother than destruction.

Read Table 1:

- Table 1 regresses job creation and job destruction on leads, contemporaneous values, and lags of output growth.
- In the full output-growth block, the sum of creation coefficients is 0.218, while the sum of destruction coefficients is −0.384.
- In expansions, creation is more responsive than destruction: the sums are 0.399 for creation and −0.066 for destruction.
- In contractions, destruction is much more responsive: the sums are 0.084 for creation and −0.634 for destruction.

Economic message: The data support the key asymmetry. Creation reacts more gently and symmetrically, while destruction is the margin that moves sharply in recessions.

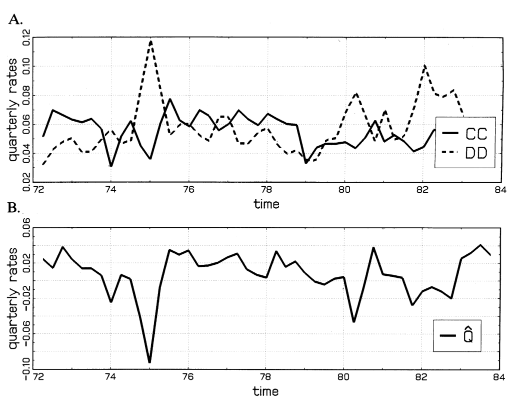

Figure 3: U.S. manufacturing job creation, destruction, and output growth.

TABLE 1—JOB CREATION AND DESTRUCTION RESPONSE TO OUTPUT GROWTH

		Creation		Destruction	
Regressor	Timing	Coefficient	Standard deviation	Coefficient	Standard deviation
$\hat{Q}$	2 leads	0.029	0.006	0.030	0.010
	1 lead	0.065	0.007	−0.068	0.010
	contemporaneous	0.108	0.007	−0.185	0.010
	1 lag	0.013	0.007	−0.103	0.010
	2 lags	0.003	0.006	−0.058	0.010
	Sum:	0.218	0.013	−0.384	0.017
$\hat{Q}^+$	2 leads	0.052	0.012	0.012	0.016
	1 lead	0.102	0.012	0.002	0.016
	contemporaneous	0.131	0.012	−0.065	0.016
	1 lag	0.059	0.012	−0.025	0.016
	2 lags	0.055	0.012	−0.008	0.016
	Sum:	0.399	0.026	−0.066	0.023
$\hat{Q}^-$	2 leads	0.002	0.010	0.006	0.014
	1 lead	0.022	0.011	−0.149	0.014
	contemporaneous	0.093	0.012	−0.293	0.015
	1 lag	−0.012	0.012	−0.139	0.015
	2 lags	−0.021	0.012	−0.059	0.015
	Sum:	0.084	0.020	−0.634	0.024

Table 1: Job creation and destruction response to output growth.

6.4 Figure 4 and Table 2: Asymmetric demand and calibration parameters

Read Figure 4:

- Figure 4 simulates the model under an asymmetric demand process.
- Demand downturns are shorter and sharper than expansions.

- Creation remains relatively smooth because it is based on expected future conditions.
- Destruction becomes highly asymmetric because it reacts to current profitability and must partly "make up" for the smoothness of creation.

Read Table 2:

- The calibration uses an interest rate of 0.065, depreciation/failure rate of 0.150, and technical progress rate of 0.028.
- The adjustment-cost parameters are $c_0 = 0.403$ and $c_1 = 0.500$.
- The value of c_1 implies only modest convexity in creation costs, but enough to break full insulation and generate cyclical destruction.

Economic message: A small amount of creation smoothing is enough to produce large destruction asymmetry. This is the core quantitative success of the model.

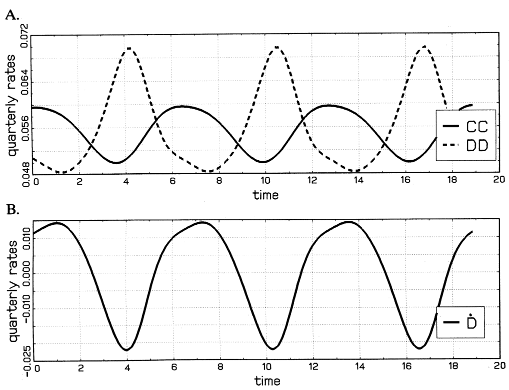

Figure 4: Creation, destruction, and asymmetric demand.

TABLE 2—CALIBRATED PARAMETERS

Variable	Symbol	Value
Interest rate	r	0.065
Depreciation rate	δ	0.150
Rate of technical progress	γ	0.028
Adjustment cost parameters	c_0	0.403
	c_1	0.500

Table 2: Calibrated parameters.

6.5 Figures 5 and 6: Model-generated job flows versus data

Read Figure 5:

- Figure 5 compares employment-driven model simulations with actual job creation and destruction.
- The model-generated destruction series captures the major spikes in the data.
- Creation is smoother in the model than in the data, partly because the model does not include uncertainty or seasonal disturbances.

Read Figure 6:

- Figure 6 compares output-driven model simulations with actual job creation and destruction.
- The model again captures the broad difference between smoother creation and more volatile destruction.

- The output-driven simulation cannot capture seasonal movements in the job-flow data because industrial production is seasonally adjusted.

Economic message: The calibrated model is deliberately simple, but it matches the main qualitative facts: destruction is more volatile and more recession-sensitive than creation.

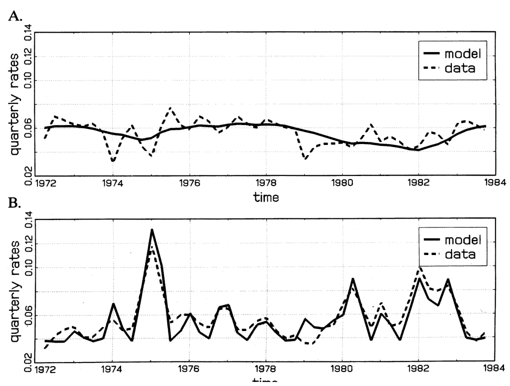

Figure 5: Employment-driven job creation and destruction.

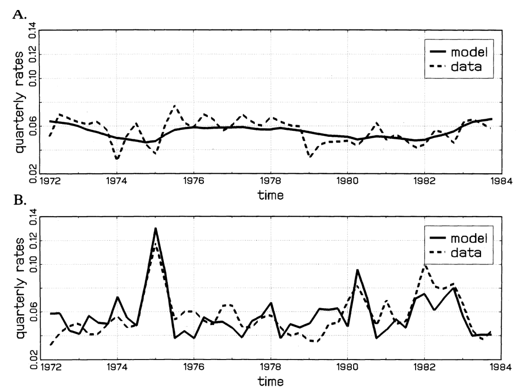

Figure 6: Output-driven job creation and destruction.

7 Short summary

- The paper studies business cycles in a vintage model of creative destruction.
- Production units embody different vintages of technology; new units are created and old units are destroyed.
- Demand fluctuations can be absorbed on either the creation margin or the destruction margin.
- If creation costs are linear, creation fully insulates existing units from demand shocks and destruction does not respond directly to the cycle.
- If creation is costly to adjust quickly, firms smooth creation and recessions raise destruction.
- U.S. manufacturing job-flow data show that job destruction is much more cyclical than job creation.
- Creation responds more to expansions than contractions, while destruction responds much more to contractions than expansions.
- The model explains this pattern because creation is forward looking and smoothed, while destruction is driven by current profitability.
- A calibrated model matches the relative volatilities and asymmetries of job creation and destruction.
- Recessions can therefore be interpreted as periods of cleansing, when old or unprofitable production units are pruned from the economy.

8 Conclusion

8.1 Core conclusion

Caballero and Hammour show that recessions can have a cleansing effect in an economy with ongoing creative destruction. Because production units differ by vintage, downturns do not affect all units equally. Older, less productive units are closer to the profitability margin and are more likely to be destroyed when demand falls.

8.2 Economic intuition

The central intuition is the interaction between creation and destruction. If firms can easily vary creation, then lower creation absorbs recessions and existing units are insulated. If creation is costly to adjust, firms smooth creation, so old units bear more of the shock. This makes job destruction highly cyclical and asymmetric.

8.3 Empirical interpretation

The U.S. manufacturing job-flow evidence fits the model well. Job creation is relatively smooth and symmetric. Job destruction is volatile and spikes during contractions. These facts are hard to understand in a representative-firm model but natural in a vintage model with heterogeneous units.

8.4 Implications

The paper implies that recessions change the composition of production, not just the level of activity. Downturns can accelerate the exit of outdated techniques and products. This provides one channel through which business cycles interact with long-run productivity dynamics.

8.5 Limitations

The paper does not show that recessions are welfare improving. Destroying obsolete units may raise average efficiency, but recessions also impose unemployment, lost output, and other costs. The paper's contribution is to model and measure a positive cleansing mechanism, not to advocate recessions as policy.

8.6 Takeaway

The enduring takeaway is that gross flows matter. A recession is not simply a fall in aggregate employment or output. It is also a reallocation episode in which creation is delayed and destruction is concentrated among older, less productive units.